Alejandro Fontal, PhD

DATA SCIENCE | MACHINE LEARNING | HEALTH & LIFE SCIENCES

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Data scientist with a PhD and 8+ years of industry and research experience in health and life sciences, currently leading the data science strategy in a MedTech R&D team. I approach data as a scientist, with clear questions, careful study and validation design, and reproducible statistical and ML workflows.

WORKING EXPERIENCE

R&D @ NIMBLE Diagnostics

Senior Machine Learning Engineer

Jul. 2026 - Present Barcelona, ES

- Lead the data science strategy in R&D for a MedTech team developing diagnostic methods from biomedical signals collected in experimental studies and clinical trials.
- Develop statistical and ML methods to extract diagnostic information from complex biomedical signals, with validation built around clinically meaningful endpoints.
- Work with clinical, signal-processing, and engineering teams on signal representation, validation design, and performance assessment.

Climate & Health Program @ ISGlobal

Postdoctoral Researcher | Data & Software Lead

Oct. 2024 - Apr. 2026 Barcelona, ES

- Led software and data engineering for the group, building internal packages, analytical tools, data pipelines, and GitHub-based workflows for reproducible research.
- Built reproducible pipelines for long-read aerobiome metagenomics and linked results to health outcomes.
- Developed supervised ML models for bacterial discrimination from laser-induced fluorescence and light-scattering measurements.

PhD Fellow | Marie Skłodowska-Curie ESR

Oct. 2019 - Sep. 2024

- Developed time-series methods to quantify environmental signals in epidemics, including transient and lagged associations, multi-scale correlations, and phase-shift analyses.
- Integrated nationwide case data with climate reanalysis, remote sensing, GIS layers, and air-mass back-trajectories to model drivers of Kawasaki Disease, COVID-19, and influenza.
- Analyzed registry and emergency-health datasets, assessing data quality, population denominators, and spatiotemporal incidence.

Protein Engineering @ DuPont Industrial Biosciences

Data Scientist

Sep. 2018 - Sep. 2019 Leiden, NL

- Built and benchmarked ML models linking protein sequence, structure, and experimental performance for protein-engineering R&D.
- Built and maintained reproducible bioinformatics and data pipelines.
- Partnered with wet-lab teams on experimental design and the model-experiment loop.

EDUCATION

PhD in Biotechnology — 2019-2024

Universitat de Barcelona

- PhD focused on environmental determinants of disease onset, spatiotemporal modeling, and long-read aerobiome metagenomics.

MSc in Bioinformatics — 2016-2018

Wageningen University & Research

- Minor in Data Science, cum laude distinction.
- Thesis on interpretable deep learning for protein subcellular localization.

BSc in Biotechnology — 2011-2015

Universitat de Barcelona

- Thesis at the VHIR Bioinformatics Unit on meta-analysis of transcriptomics tools.

Languages: Spanish & Catalan (native), English (C2)

TECHNICAL SKILLS

Statistics & Data Analysis: Statistical modeling and inference, time-series and spatiotemporal analysis, lagged associations, count and longitudinal data, and validation design.

Health & Biomedical Data: Clinical-trial, registry, surveillance, and emergency-health data, and biomedical signals; incidence, denominators, and data quality.

Python: Expert (9+ years). pandas, polars, statsmodels, scikit-learn, PyTorch, TensorFlow, and plotnine; package development and testing.

R: Tidyverse, Bioconductor, epidemiological analysis, visualization, reproducible reports, and R/Python workflows.

Data Engineering: SQL/DuckDB, ETL pipelines, Parquet/Arrow, and Jupyter/Quarto reports.

Git & DevOps: Collaborative development, code review, and automated testing and deployment (GitHub Actions, GitLab CI/CD).

Linux & HPC: Daily Linux user; shell scripting, HPC workflows (SLURM, Nextflow), containers, and remote deployment.

AI/Agentic Development: Tool-using coding and automation workflows with Codex, Claude Code, and OpenCode.